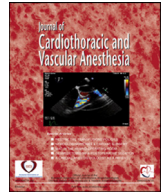




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Original Article

The Clinical Application of Ultra-Fast-Track Cardiac Anesthesia in Right-Thoracoscopic Minimally Invasive Cardiac Surgery: A Retrospective Observational Study

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Objectives: The purpose of this study was to investigate the effect of ultra-fast-track cardiac anesthesia (UFTCA) on rapid postoperative recovery in patients undergoing right-thoracoscopic minimally invasive cardiac surgery.

Design: A retrospective observational study.

Setting: A single large teaching hospital.

Participants: A total of 153 patients who underwent right-thoracoscopic minimally invasive cardiac surgery between January 2021 and August 2021 were enrolled. The inclusion criteria were American Society of Anesthesiologists grade I to III, New York Heart Association (NYHA) cardiac function class I to III, and age ≥ 18 years. The exclusion criteria were NYHA class IV, local anesthetic allergy, severe pulmonary hypertension (pulmonary arterial systolic pressure, PASP > 70 mmHg), age ≤ 18 years or ≥ 80 years old, emergency surgery, and patients with incomplete or missing data.

Interventions: Finally, a total of 122 patients were included and grouped by different anesthesia strategies. Sixty patients received serratus anterior plane block-assisted ultra-fast-track cardiac anesthesia (UFTCA group), and 62 patients received conventional general anesthesia (CGA group). The primary outcomes were lengths of hospital stay and postoperative intensive care unit (ICU) stay. The secondary outcomes were postoperative pain scores, opioids use, postoperative chest tube drainage, and complications.

Measurements and Main Results: The intraoperative dosages of sufentanil and remifentanyl in the UFTCA group were significantly lower than those in the CGA group ($66.25 \pm 1.03 \mu\text{g}$ v $283.31 \pm 11.36 \mu\text{g}$, $p < 0.001$; and $1.94 \pm 0.38 \text{ mg}$ v $2.14 \pm 0.99 \text{ mg}$, $p < 0.001$, respectively). The incidence of postoperative rescue analgesia in the UFTCA group was significantly lower than that in the CGA group (10 patients [16.67%] v 30 patients [48.38%], $p < 0.001$). In the postoperative ICU, there were fewer patients with pain score Numeric Rating Scale ≥ 3 in the UFTCA group than that in the CGA group (10 patients [16.67%] v 29 patients [46.78%], $p < 0.001$). The postoperative extubation time in the UFTCA group was shorter than that in the CGA group (0.3 hours [range, 0.25–0.4 hours] v 13.84 hours [range, 10.25–18.36 hours], $p < 0.001$). Lengths of ICU stay and hospital stay in the UFTCA group were shorter than those in the CGA group (27.73 ± 16.54 hours v 61.69 ± 32.48 hours, $p < 0.001$; and 8 days [range, 7–9] v 9 days [range, 8–12], $p < 0.001$, respectively). Compared with the CGA group, the patients in the UFTCA group had less chest tube drainage within 24 hours after surgery ($197.67 \pm 13.05 \text{ mL}$ v $318.23 \pm 160.10 \text{ mL}$, $p < 0.001$). There were no significant differences in in-hospital mortality, postoperative bleeding, or secondary surgery between the 2 groups. The incidences of postoperative nausea, vomiting, or atelectasis were comparable between the 2 groups.

Conclusions: Serratus anterior plane block-assisted ultra-fast-track cardiac anesthesia can promote rapid postoperative recovery in patients with right-thoracoscopic minimally invasive cardiac surgery. This anesthesia regimen is clinically safe and feasible.

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Key Words: ultra-fast track cardiac anesthesia; minimally invasive cardiac surgery; serratus anterior plane block; thoracoscope

WITH THE ADVANCEMENT of surgical techniques, the improvement of cardiopulmonary bypass (CPB), and the application of minimally invasive thoracoscopy, the trauma of cardiovascular surgery has been reduced greatly, and postoperative recovery has been improved.^{1,2} Thoracoscopic right-side minimally invasive cardiac surgery has emerged as an alternative to conventional cardiac surgery, and is increasingly accepted by patients based on favorable short- and long-term outcomes.

The concept of enhanced recovery after cardiac surgery³ was proposed on the basis of “fast-track” and “ultra-fast-track” cardiac anesthesia (FTCA and UFTCA). The joint efforts of medical teams, such as cardiac surgery, anesthesia, perfusion, and intensive care, are the basis for promoting accelerated recovery of patients undergoing cardiovascular surgery.^{4,5} Fast-track cardiac anesthesia refers to a patient’s recovery from anesthesia and extubation with an appropriate anesthetic regimen within 6 hours after surgery. Ultra-fast-track cardiac anesthesia means that patients recover and are extubated within 1 hour, even immediately after surgery. The purpose of FTCA and UFTCA is to reduce intensive care unit (ICU) and hospital stay and associated complications, ultimately promoting recovery and reducing postoperative healthcare costs.⁶⁻⁸

Since January 2021, the authors have performed a serratus anterior plane block (SAPB) to assist UFTCA. The authors conducted this observational study under the assumption that UFTCA in patients undergoing minimally invasive cardiovascular surgery would promote rapid postoperative recovery. The authors’ primary endpoints were lengths of postoperative hospital stay and ICU stay, and the secondary endpoints were postoperative pain scores, opioid use, postoperative chest tube drainage, and complications.

Materials and Methods

Patient Population

The retrospective study enrolled 153 patients who underwent right-thoracoscopic minimally invasive cardiac surgery in the authors’ cardiac center from January 2021 to August 2021. The Institutional Review Board (IRB) of Zhejiang Provincial People’s Hospital (IRB No.2021KY045; approval date: October 18, 2021) approved this study. The IRB waived informed consent due to the retrospective nature of the analysis. The inclusion criteria were American Society of Anesthesiologists class I to III, New York Heart Association (NYHA) cardiac function class I to III, and age ≥ 18 years. The exclusion criteria were NYHA class IV, local anesthetic allergy, severe pulmonary hypertension (pulmonary arterial systolic pressure, PASP > 70 mmHg), age ≤ 18 years or ≥ 80 years, emergency surgery, and patients with incomplete or missing data.

General Data

Among the 153 patients, 31 patients were excluded, including 10 patients with NYHA class IV, 17 patients whose age did not meet the inclusion criteria, and 4 patients whose medical records were incomplete. Ultimately, 122 patients were eligible for analysis, with 27 undergoing mitral valve replacement, 38 mitral valve repair, 26 aortic valve replacement, and 31 aortic valve replacement and mitral valve replacement. Of the 122 patients, 60 patients received SAPB-assisted UFTCA (UFTCA group) and 62 received CGA (CGA group) (Fig 1).

Patients in the UFTCA Group

Ultrasound-guided SAPB: To improve patient comfort, SAPB was performed after anesthesia induction, using a high-frequency ultrasound probe to locate the fourth rib in the right midaxillary line. Ropivacaine (20 mL, 0.375%) was injected into the deep surface of the serratus anterior muscle using the in-plane needle technique. Ultrasound-guided SAPBs were performed by the same anesthesiologist (Fig 2).

Anesthesia Induction: Anesthesia was induced with 0.2 mg/kg of etomidate, 0.5-to-1.0 mg/kg of propofol, 0.4-to-0.5 μ g/kg of sufentanil, and 0.15 mg/kg of benzenesulfonate atracurium.

Anesthesia was maintained with an intravenous infusion of 0.1-to-0.5 μ g/kg/min of remifentanyl and 100-to-200 μ g/kg/min of propofol. Benzenesulfonate atracurium, 1.0-to-2.0 μ g/kg/min, was intravenously infused. The routine dosage of sufentanil is 0.2-to-0.5 μ g/kg before incision, 0.2-to-0.3 μ g/kg at the beginning of CPB, 0.2-to-0.3 μ g/kg after aortic clamping. The total amount of sufentanil during the surgery was controlled at 1-to-1.5 μ g/kg. The dosages of remifentanyl and propofol were adjusted according to the degree of surgical stimulation to maintain the bispectral (BIS) value between 30 and 40 (the use of benzodiazepine sedatives was prohibited during anesthesia).

Recovery From Anesthesia: One hour before the end of the surgery, the infusion of propofol was reduced according to the degree of surgical stimulation, and the infusion of benzenesulfonate atracurium was stopped. All intravenous anesthetics were discontinued 30 minutes before the end of surgery. If necessary, 20-to-30 mg of propofol were injected to keep the patient sedated until the end of surgery. The tracheal tube was removed in patients who were awake immediately after surgery. When extubation was not possible within 30 minutes after surgery, the patient was transferred to the post-anesthesia care unit. If the tracheal tube could not be removed within 1 hour in the post-anesthesia care unit, the UFTCA was abandoned, and the patient was transferred to the ICU with the tracheal tube.

Patients were extubated if the following criteria were met: (1) hemodynamically stable, only small doses of vasoactive drugs

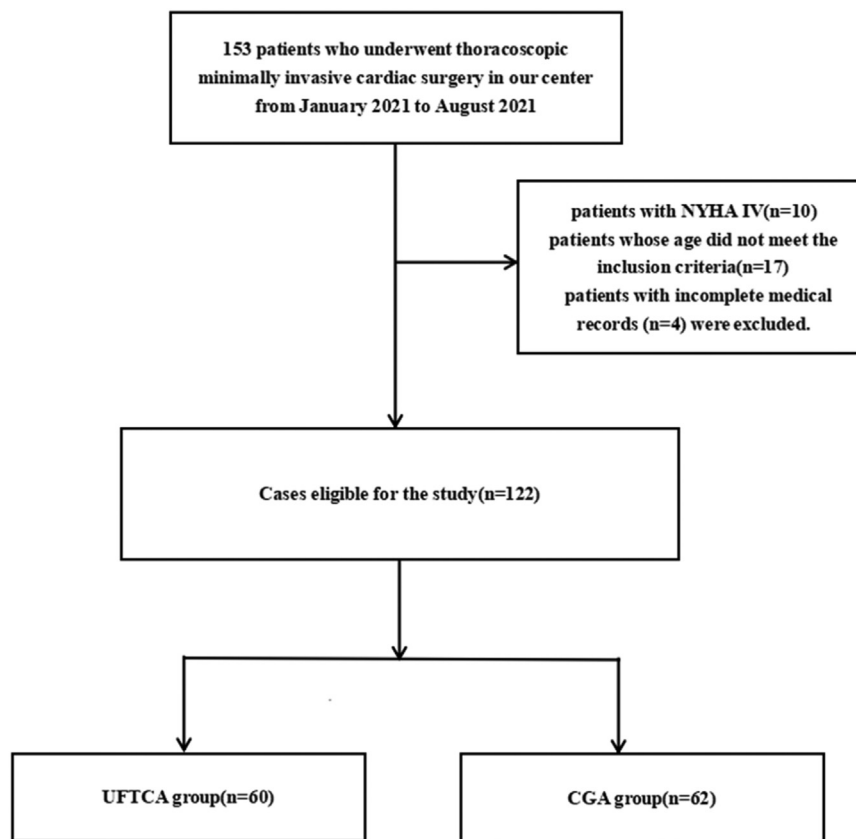


Fig 1. Study flowchart. CGA, conventional general anesthesia; UFTCA, ultra-fast-track cardiac anesthesia.

were needed, such as dobutamine $\leq 5.0 \mu\text{g}/(\text{kg}/\text{min})$ and norepinephrine $\leq 0.05 \mu\text{g}/(\text{kg}/\text{min})$; (2) hemoglobin $\geq 80 \text{ g/L}$; (3) hydrogen ion concentration(PH) value was maintained at 7.35-to-7.45; (4) no active bleeding; (5) restoration of consciousness, such as opening and closing eyes, sticking out tongue, and gripping and releasing hands as commanded; (6) spontaneous breathing was smooth, tidal volume returned to normal, and airway defense reflexes recovered; and (7) pulse oxygen saturation $\geq 95\%$. Neuromuscular blockade was reversed with neostigmine if needed.

Patients in the CGA Group

Anesthesia Induction: Anesthesia was induced with 0.2 mg/kg of etomidate, $0.5\text{-}1.0 \text{ mg/kg}$ of propofol, $0.4\text{-}0.5 \mu\text{g/kg}$ of sufentanil, and 0.15 mg/kg of benzenesulfonate atracurium.

Anesthesia was maintained with an intravenous infusion of $0.1\text{-}0.5 \mu\text{g/kg}/\text{min}$ remifentanyl, $100\text{-}200 \mu\text{g/kg}/\text{min}$ propofol, and $1.0\text{-}2.0 \mu\text{g/kg}/\text{min}$ of benzenesulfonate atracurium. The conventional cardiac anesthesia is total intravenous anesthesia in the authors' center, including $0.5\text{-}1 \mu\text{g/kg}$ of sufentanil before incision, $0.5\text{-}1 \mu\text{g/kg}$ at the beginning of CPB, $0.5\text{-}1 \mu\text{g/kg}$ after aortic clamping, and $0.5\text{-}1 \mu\text{g/kg}$ at the end of CPB. The total intraoperative amount of sufentanil was approximately $5 \mu\text{g/kg}$ according to the length of surgery. The dosages of remifentanyl, propofol, and sufentanil were adjusted according to the degree of surgical stimulation to maintain the BIS value between 30 and 40 (benzodiazepine sedatives were prohibited during anesthesia). After the surgery, the patient immediately was transferred to the ICU for rehabilitation and extubation.

All patients were connected to the intravenous analgesia pumps in the ICU. If the intravenous analgesia pump reached the limit dose of sufentanil, the Numeric Rating Scale (NRS) visual analog pain score was still ≥ 3 points, and the Critical Care Pain Observation Tool (CPOT) score was ≥ 3 points, intravenous rescue analgesia was given ($100 \mu\text{g/mL}$ of remifentanyl intravenous micropump infusion, $0.5\text{-}1 \mu\text{g/kg}/\text{min}$). Patients were transferred to the general ward in stable condition. If the pain score in the general ward was ≥ 3 , 0.1 g of tramadol was used for rescue analgesia.

Surgical Procedure

The patient was placed in the supine position, and a transthoracic echocardiography probe was inserted. As reported

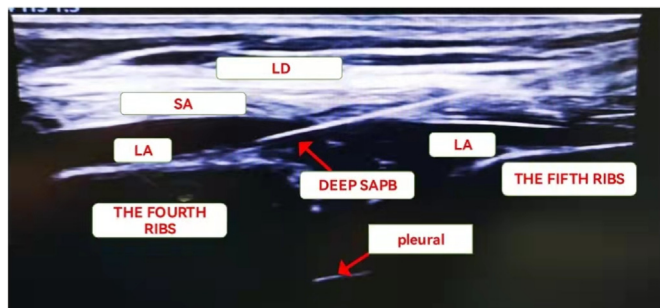


Fig 2. Ultrasound-guided serratus anterior plane block: sonographic landmarks. LA, local anesthetic; LD, latissimus dorsi; SA, serratus anterior muscle; SAPB, serratus anterior plane block.

previously,⁹ a 4-cm incision was made through the right thoracic fourth intercostal space and a rib spreader was placed. The pericardium was dissected and suspended before surgery.

Data Extraction and Collation

Clinical data were extracted from the electronic medical record system, including age, sex, body mass index, comorbidities (eg, hypertension, diabetes, atrial fibrillation, coronary heart disease, and history of cerebral infarction), lung function, surgical method, aortic clamp time, CPB time, surgery time, and anesthesia method.

Outcome Indicators

The outcome indicators included the incidence of postoperative rescue analgesia, the incidence of CPOT ≥ 3 or NRS ≥ 3 in the ICU and general ward, postoperative mechanical ventilation time, lengths of ICU stay and hospital stay, postoperative chest tube drainage within the first 24 hours, nausea and vomiting, atelectasis, in-hospital mortality, and incidence of postoperative bleeding and secondary surgery. In this observation, atelectasis was only a qualitative indicator based on the diagnostic report, and the severity was not graded.

Statistical Analysis

The mean difference in postoperative hospital stay between the 2 groups was 2.5 days, and the common SD of the 2 groups

was 4 days. The difference test was adopted, with $\beta = 0.2$ and $\alpha = 0.05$ (bilateral test), and the ratio of the 2 groups was set as 1:1. SPSS 15 was used to calculate the sample size, with a minimum of 42 patients per group. Considering a 15% drop, a total of 122 patients were included in the study. The SPSS, version 25 statistical software (IBM SPSS, Inc) was used for analysis. The continuous variables with normal distribution were expressed as mean $\pm$ SD ($\bar{x} \pm s$), and skewed data were presented as median (IQR). The quantitative variables were compared using *t*-test, and categorical variables were compared using χ^2 test or Fisher's exact test. A *p* value < 0.05 was considered statistically significant.

Results

All 60 patients in the UFTCA group successfully underwent UFTCA as planned. There were no significant differences in demographic characteristics, surgery time, CPB, and aortic clamping, type of surgery, or intraoperative blood transfusion. The intraoperative doses of sufentanil and remifentanyl in the UFTCA group were significantly lower than those in the CGA group (sufentanil, $66.25 \pm 1.03 \mu\text{g}$ v $283.31 \pm 11.36 \mu\text{g}$, $p < 0.001$; remifentanyl, $1.94 \pm 0.38 \text{ mg}$ v $2.14 \pm 0.99 \text{ mg}$, $p < 0.001$, respectively) (Table 1).

The incidences of postoperative ICU rescue analgesia were 10 (16.67%) in the UFTCA group, and 30 (48.38%) in the CGA group ($p < 0.001$). The incidences of postoperative rescue analgesia in the general wards were 15 (25.0%) in the

Table 1

The basic demographic characteristics, operation related times and incidence of intraoperative blood transfusion in the patient cohort.

	Overall (n=122)	UFTCA group (n=60)	CGA group(n=62)	P value
Age(y)	56.76 $\pm$ 12.21	57.68 $\pm$ 12.55	55.87 $\pm$ 11.907	0.415
Gender (male/female)	(65/57)	(31/29)	(34/28)	0.726
Height (cm)	162.95 $\pm$ 8.63	163.05 $\pm$ 8.17	162.85 $\pm$ 9.13	0.901
Weight (kg)	61.42 $\pm$ 11.77	60.53 $\pm$ 10.62	62.29 $\pm$ 12.81	0.413
BMI (kg/m ²)	23.12 $\pm$ 3.35	22.66 $\pm$ 2.88	23.57 $\pm$ 3.73	0.138
ASA	3(3-3)	3(3-3)	3(3-3)	0.368
NYHA	3(2-3)	2(2-3)	2.5(2-3)	0.292
Comorbidity				
Hypertension(n/%)	35(28.7)	17(28.3)	18(29.0)	0.932
Diabetes(n/%)	11(9.0)	5(8.3)	6(9.7)	0.796
Atrial fibrillation(n/%)	22(18.0)	7(11.7)	15(24.2)	0.072
Coronary heart disease(n/%)	12(9.8)	4(10.0)	6(9.7)	0.952
History of cerebral infarction(n/%)	5(4.1)	2(3.3)	3(4.8)	1.000
Type of surgery				
MVR(n/%)	27 (22.13)	14 (23.33)	13 (20.97)	
MVP(n/%)	35 (28.69)	18 (30.0)	17 (27.42)	
AVR(n/%)	26 (21.31)	13 (21.67)	13 (20.97)	
AVR and MVR(n/%)	31 (25.41)	15 (25.0)	16 (25.81)	
Operation (min)	281.97 $\pm$ 68.25	270.38 $\pm$ 7.941	293.63 $\pm$ 78.16	0.064
CPB time (min)	139.39 $\pm$ 28.55	143.77 $\pm$ 25.74	135.15 $\pm$ 30.65	0.141
Aortic clamp time (min)	88.73 $\pm$ 28.89	93.13 $\pm$ 24.68	84.47 $\pm$ 32.07	0.544
Intraoperative blood transfusion	22(18.03)	10(16.67)	12(19.35)	0.699

NOTE. Continuous data were represented as mean SD, categorical data by percentage (%), and ASA and NYHA by median (IQR). Statistically significant difference ($p < 0.05$).

Abbreviations: ASA, American Society of Anesthesiologists; BMI, body mass index; AVR, aortic valve replacement; CGA, conventional general anesthesia; CPB, cardiopulmonary bypass; NYHA, New York Heart Association; MVP, mitral valve repair; MVR, mitral valve replacement; UFTCA, ultrafast-track cardiac surgery anesthesia.

Table 2
Intraoperative Amount of Opioids and Postoperative Analgesia

	Overall (N = 122)	UFTCA Group (n = 60)	CGA Group (n = 62)	p Value
Sufentanil intraoperatively (ug)	185.52 ± 140.97	67.0 ± 8.55	300.24 ± 110.29	< 0.001*
Semifentanil intraoperatively (mg)	2.04 ± 0.76	1.94 ± 0.38	2.14 ± 0.99	< 0.001*
ICU rescue analgesia (n/%)	40 (32.79)	10 (17.67)	30 (48.38)	< 0.001*
Ward rescue analgesia (n/%)	27 (22.13)	15 (25.0)	12 (19.35)	0.453

NOTE. The continuous data were expressed as the mean ± SD, and categorical data were expressed as percentage (%).

Abbreviations: CGA, conventional general anesthesia; ICU, intensive care unit; UFTCA, ultrafast track cardiac surgery anesthesia.

* Significant difference.

UFTCA group and 12 (19.35%) in the CGA group ($p = 0.453$). There was no significant difference in postoperative rescue analgesia in the general ward (Table 2).

Postoperative ICU NRS ≥ 3 was in 10 patients (16.67%) in the UFTCA group and 29 (46.78%) in CGA group ($p < 0.001$). Ward NRS ≥ 3 in UFTCA group was in 15 patients (25.0%) and 8 (12.90%) in CGA group ($p = 0.088$) (Table 3). Postoperative ICU CPOT ≥ 3 was in 29 patients (46.78%) in the CGA group. Three patients in the CGA group switched to rescue analgesia due to side effects from the use of intravenous analgesic pump.

All patients in UFTCA group were extubated immediately, or within 1 hour after surgery (extubation time range from 0.25–0.4 hours), whereas patients in CGA group were extubated in the ICU (extubation time range from 10.25–18.36 hours) ($p < 0.001$). The postoperative ICU stay, hospital stay, and 24-hour postoperative chest tube drainage were significantly lower in the UFTCA group than in the CGA group ($p < 0.001$) (Table 4).

There were no statistical differences in the incidences of postoperative complications, bleeding, secondary surgery, or in-hospital mortality between the 2 groups (Table 5).

Discussion

It is imperative to improve the quality of cardiovascular surgery and promote rapid postoperative recovery,¹⁰ which is a constant concern of cardiac surgeons and anesthesiologists. Since 2018, thoracoscopy-assisted right infraaxillary minimally invasive incision cardiac surgery has become a routine procedure in the authors' cardiac center. A variety of procedures can be performed under this minimally invasive approach, including valve replacement or repair, repair of

complex congenital heart disease, and septal myectomy for hypertrophic obstructive cardiomyopathy.⁹ Given the growing interest in FTCA and UFTCA for cardiac surgery, the authors innovatively performed UFTCA in some patients undergoing minimally invasive cardiac surgery. In this retrospective observation, the authors compared UFTCA with CGA, and determined that UFTCA strategy is beneficial for better postoperative recovery.

Patients in the CGA group were administered more sufentanil and remifentanil and were transferred to the ICU for anesthesia recovery and extubation. Traditional cardiac anesthesia with high-dose opioids had obvious analgesic effects, with mild myocardial injury and hemodynamic influence. It can significantly reduce the cardiovascular response to tracheal intubation and surgical stimulation without the release of histamine, which is called "stress-free anesthesia."¹¹ That was the reason for the authors' previously large amounts of sufentanil and remifentanil.

The implementation of enhanced recovery after cardiac surgery means that cardiac anesthesia needs to replace the previous application of high-dose opioids with a new perioperative multimodal analgesia technique; that is, reduced opioids or opioid-free⁸ combined with balanced short-acting anesthetics. In the authors' UFTCA strategy, total intravenous anesthesia combined with SAPB were used, and the total intraoperative dose of sufentanil was controlled at 1-to-1.5 $\mu\text{g/kg}$.

Ultrasound-guided right deep SAPB was used for UFTCA. Injection of local anesthetic between the fourth rib serratus and the surface of the rib (deep SAPB) effectively blocked pain conduction from the lateral cutaneous branch of the T2–6 intercostal nerves.¹² Thus, deep SAPB as an effective analgesic on the right chest wall^{13,14} reduced intraoperative opioids use, reduced postoperative pain scores, and reduced the

Table 3
Incidence of CPOT and NRS Score ≥ 3 in the ICU and Ward

	Overall (N = 122)	UFTCA Group (n = 60)	CGA Group (n = 62)	p Value
CPOT ³ in ICU (n/%)	29 (23.78)		29 (46.78)	
NRS ³ in ICU (n/%)	39 (31.97)	10 (16.67)	29 (46.78)	< 0.001*
NRS ³ in Ward (n/%)	23 (18.85)	15 (25.0)	8 (12.90)	0.088

NOTE. The categorical variables were expressed as percentages (%).

Abbreviations: CGA, conventional general anesthesia; CPOT, intensive care patient pain score; ICU, intensive care unit; NRS, Numerical Rating Scale pain score; UFTCA, ultrafast-track cardiac surgery anesthesia.

Table 4
Postoperative Recovery Time and Chest Tube Drainage

	Overall (N = 122)	UFTCA Group (n = 60)	CGA Group (n = 62)	p Value
Postoperative extubation time (h)	6.64 (0.30-13.94)	0.3 (0.25-0.4)	13.84 (10.25-18.36)	< 0.001*
Length of stay in ICU (h)	44.99 ± 30.92	27.73 ± 16.54	61.69 ± 32.48	< 0.001*
Length of stay in hospital (d)	8 (7-10)	8 (7-9)	9 (8-12)	< 0.001*
Chest tube drainage (mL)	258.93 ± 159.19	197.67 ± 130	318.23 ± 160.10	< 0.05

NOTE. Continuous data were represented as the mean ± SD, and postoperative extubation time was represented as median (IQR).

Abbreviations: CGA, conventional general anesthesia; ICU, intensive care unit; UFTCA, ultrafast-track cardiac surgery anesthesia.

* Significant difference.

incidence of ICU rescue analgesia. Serratus anterior plane block has been one of the assistive techniques for UFTCA since January 2021 for right-thoroscopic minimally invasive cardiac surgery in the authors' center. Currently, SAPB, combined with pectoralis I and pectoralis II block, have been routinely applied in all patients undergoing right-thoroscopic minimally invasive cardiac surgery in the authors' center, regardless of UFTCA or not.

Good perioperative analgesia management is the basis for successful UFTCA. Although the intraoperative opioids consumption in the UFTCA group was lower than that in the CGA group, the results suggested that the anesthesia regimen of SAPB-assisted UFTCA could achieve better postoperative analgesia. Fewer patients needed rescue postoperative analgesia in the ICU in the UFTCA group. The incidence of NRS ≥ 3 in the postoperative ICU in the UFTCA group was significantly lower than that in the CGA group. However, a single injection of 20 mL of 0.375% ropivacaine for SAPB did not provide prolonged analgesia. As the block effect subsided, there was no significant difference in analgesia between the 2 groups in the general ward (Table 2). Notably, the use of high-dose opioid analgesics resulted in delayed extubation, significantly prolonged postoperative mechanical ventilation, and hindered postoperative recovery.^{15,16}

Patients in the UFTCA group had shorter extubation time and shorter ICU stay. Consistent with previous studies, the authors' UFTCA also reduced the hospital stay.^{17,18} Studies showed that extubation in the operating room for patients with congenital heart disease was safe and feasible.¹⁹ Others proved that early extubation could reduce the dependence of patients on vasoactive drugs after surgery. It also could reduce

postoperative fluid and blood product transfusion.²⁰ Accelerating ICU discharge is pivotal for rapid postoperative recovery, speeding up ICU bed turnover, and alleviating the shortage of medical resources. The prolonged mechanical ventilation time in the CGA group was related partially to the previous shortage of ICU physicians and the avoidance of extubation at night for safety reasons. Ultra-fast-track cardiac anesthesia allowed patients to resume spontaneous ventilation as soon as possible after surgery, minimizing mechanical ventilation and length of stay in the ICU. This was what motivated the authors to conduct UFTCA.

Another interesting finding was that patients in the UFTCA group had less thoracic chest tube drainage within 24 hours after surgery. The specific mechanism was unclear. Less postoperative blood loss can lead to less postoperative blood transfusion, thereby reducing the incidence of transfusion-related complications.¹⁵ Although there was no statistically significant difference between the 2 groups, a subset of patients in both groups developed postoperative atelectasis, which deserves clinical attention.

The authors did not observe differences of postoperative nausea or vomiting between the 2 groups, which was contrary to the findings of Park et al.²¹ That may have been due to the routine use of 10 mg of dexamethasone or 5 mg/kg of methylprednisolone during the surgery, the antiemetic drugs for all cardiac surgery patients in the authors' center. Studies have shown that intraoperative glucocorticoids administration can prevent postoperative nausea and vomiting.²² One patient in the CGA group underwent reoperation to stop active bleeding from the chest wall. There were no SAPB-related complications in the UFTCA group, such as local anesthetic toxicity,

Table 5
Summary of the Postoperative Complications

	Overall (N = 122)	UFTCA Group (n = 60)	CGA Group (n = 62)	p Value
Nausea (n/%)	13 (10.66)	5 (8.33)	8 (12.90)	0.560
Vomiting (n/%)	5 (4.09)	2 (3.33)	3 (4.83)	1.000
Atelectasis (n/%)	31 (25.41)	17 (28.33)	15 (23.33)	0.535
In-hospital mortality (n/%)	0 (0)	0 (0)	0 (0)	
Postoperative hemorrhage (n/%)	1 (0.82)	0 (0)	1 (1.61)	0.608
Secondary surgery (n/%)	1 (0.82)	0 (0)	1 (1.61)	0.608

NOTE. The categorical variables were expressed as percentages (%).

Abbreviations: CGA, conventional general anesthesia; UFTCA, ultrafast-track cardiac surgery anesthesia.

hematoma, or puncture site infection. All patients recovered and were discharged. There were no significant differences between the 2 groups in in-hospital mortality, postoperative hemorrhage, and secondary surgery.

The authors maintained excessive depth of anesthesia with BIS value between 30 and 40 with continuous-infusion propofol in cardiac anesthesia before October 2021. Propofol infusing was discontinued until the end of suturing in the CGA group, whereas it was discontinued earlier in the UFTCA group. But the dosage of propofol was not counted. Since October 2021, the authors changed the unnecessary excessive anesthesia depth to maintain a BIS value between 40 and 60 in cardiac anesthesia.

There were some limitations of the current study. This study was a retrospective observation, focusing on the occurrence of adverse events during hospitalization and did not follow-up on long-term prognosis. Furthermore, the lack of randomization may have biased the results. The authors are conducting further randomized studies focusing on long-term outcomes of UFTCA.

Conclusions

Serratus anterior plane block—assisted UFTCA promotes rapid postoperative recovery in patients with right-thoracoscopic minimally invasive cardiac surgery. This approach is safe and feasible in clinical practice.

Conflict of Interest

None.

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